

ENTERPRISE IT INFRASTRUCTURE PLAN

Company Name: ApexByte Solutions Inc.
Course: ITEP 414 - System Administration and Maintenance
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PART 1: COMPANY PROFILE

ApexByte Solutions Inc. is a newly established software development firm specializing in enterprise-grade software applications, custom client integrations, and digital products. Operating from a single floor office, the company requires a secure and scalable IT foundation built from scratch.

Company Details:

- **Nature of Business:** Software Development & IT Consulting
- **Office Location:** 6th Floor, Cybertech Tower, Santa Rosa, Laguna, Philippines
- **Vision:** To become a trusted regional provider of scalable software solutions backed by reliable IT infrastructure.
- **Mission:** To deliver high-quality software products while maintaining operational efficiency and data security.

Organizational Structure & Staffing Distribution

Department	Headcount	Primary IT Functional Role
Information Technology	5	Software development, local testing, and system administration.
Human Resources	4	Personnel management, payroll processing, and internal documentation.
Finance	5	Financial auditing, accounting operations, and budget tracking.
Sales & Marketing	6	Client acquisition, product presentations, and sales operations.
Total Workforce	20	Single-floor office deployment.

PART 2: ENTERPRISE HARDWARE INVENTORY

Asset ID	Hardware Item	Qty	Department	Purpose & Technical Justification
HW-DESK-01	Developer Workstation	5	IT Department	Intel i7, 32GB RAM, 1TB SSD for software compilation and Docker testing environments.
HW-DESK-02	Office Desktop	9	HR (4), Finance (5)	Intel i5, 16GB RAM, 512GB SSD for office software, payroll, and browser applications.
HW-LAP-01	Business Laptop	6	Sales Department	Intel i5, 16GB RAM, 512GB SSD for mobile sales demos and client presentations.
HW-SRV-01	Rackmount Server	1	IT / Infrastructure	Dual Intel Xeon, 64GB ECC RAM, RAID 10 for Active Directory and local database hosting.

Asset ID	Hardware Item	Qty	Department	Purpose & Technical Justification
HW-NAS-01	NAS Storage	1	All Departments	4-Bay 16TB NAS (RAID 5) for centralized document storage and local automated backups.
HW-RTR-01	Enterprise Router	1	IT / Infrastructure	Gigabit router managing public IP routing, WAN gateway traffic, and network stability.
HW-FW-01	Hardware Firewall	1	IT / Infrastructure	Dedicated security appliance enforcing traffic rules and client VPN connections.
HW-SW-01	48-Port Managed Switch	1	IT / Infrastructure	Layer 2/3 Gigabit switch supporting VLAN configuration for departmental isolation.
HW-AP-01	Wi-Fi Access Point	2	Office Floor	Enterprise wireless access points providing Wi-Fi coverage for laptops and guest access.
HW-PRN-01	Network Printer	1	Shared Access	Networked multifunction printer/scanner for office documentation needs.
HW-UPS-01	3000VA UPS Unit	2	Server Cabinet	Uninterruptible power supply protecting core server and network hardware during power cuts.
HW-EXT-01	4TB External Hard Drive	2	IT Department	External drives used for physical offsite backups.
HW-MON-01	24" FHD Monitor	25	All Departments	Dual displays for IT developers (10 monitors) and single displays for staff (15 monitors).

PART 3: ENTERPRISE SOFTWARE INVENTORY

Software	Version	License Type	Purpose & Justification
Windows 11 Pro	23H2	Commercial	Standard OS for workstations; supports Active Directory domain join and BitLocker encryption.
Ubuntu Server	24.04 LTS	Open Source	Server OS for running local database instances, application staging, and internal tools.
Microsoft 365	Standard	Subscription	Productivity software (Word, Excel, Outlook, Teams) for daily corporate communication.
VS Code	Latest	Open Source	Standard code editor used by software developers.
Git	Latest	Open Source	Version control system for source code management.
GitHub Desktop	Latest	Open Source	Graphical client for managing Git workflows.
VirtualBox	7.0	Open Source	Hypervisor for creating test virtual machines in isolated environments.
Google Chrome	Latest	Enterprise	Primary web browser for enterprise use.
Microsoft Defender	Business	Commercial	Endpoint protection software offering antivirus and malware monitoring.

Software	Version	License Type	Purpose & Justification
AnyDesk	Enterprise	Commercial	Remote support software used by IT for internal troubleshooting.
7-Zip	Latest	Open Source	Utility for compressing and archiving files.

PART 4: ENTERPRISE NETWORK INVENTORY

Equipment	Specifications	Qty	Role in Network
ISP Fiber Modem	Enterprise Fiber ONT	1	Terminates the incoming fiber optic line from the provider.
Enterprise Router	Multi-WAN Router Gateway	1	Manages public IP traffic routing and main internet gateway functions.
Hardware Firewall	Next-Generation Firewall	1	Filters inbound and outbound traffic, blocking unauthorized connections.
Managed Switch	48-Port Gigabit Switch	1	Connects all wired devices and isolates network segments via VLANs.
Access Points	Wi-Fi 6 Enterprise AP	2	Provides wireless access for mobile laptops and guest users.
Patch Panel	48-Port CAT6 Panel	1	Organizes cable drops coming from office wall ports into the server rack.
CAT6 Cabling	UTP Copper CAT6 Cable	500m	Wired cabling connecting computers and wall jacks to the central switch.
RJ45 Connectors	Shielded RJ45 Plugs	1 Box	Connectors for terminating network cables.

PART 5: ENTERPRISE NETWORK DIAGRAM

Logical Network Architecture Sequence: [Internet] → [ISP Modem] → [Router] → [Firewall] → [Managed Switch] Switch Port & VLAN Assignments: • VLAN 10 (IT Dept): 192.168.10.0/24 • VLAN 20 (HR Dept): 192.168.20.0/24 • VLAN 30 (Finance Dept): 192.168.30.0/24 • VLAN 40 (Sales Wireless): 192.168.40.0/24 • VLAN 50 (Servers & Storage): 192.168.50.0/24

PART 6: SYSTEM ADMINISTRATION ROLES RESEARCH

1. Helpdesk Technician

- **Responsibilities:** Resolves basic user issues, sets up new computer hardware, installs software, resets passwords, and logs support tickets.
- **Skills:** OS troubleshooting, basic network tools, customer support.
- **Tools:** Jira Service Desk, AnyDesk, Active Directory Administrative Center.
- **Certifications:** CompTIA A+, Microsoft Certified: Modern Desktop Administrator.

2. Network Administrator

- **Responsibilities:** Configures routers, switches, and firewalls; monitors bandwidth; sets up VLANs; enforces security policies.
- **Skills:** IP Subnetting, routing, VLAN configuration, packet analysis.
- **Tools:** Cisco Packet Tracer, Wireshark, SolarWinds, PuTTY.
- **Certifications:** CCNA, CompTIA Network+.

3. Linux System Administrator

- **Responsibilities:** Manages Linux server environments, sets up web servers, manages file permissions, writes administrative scripts, and updates packages.
- **Skills:** Bash scripting, Linux user management, SSH, systemd services.
- **Tools:** Terminal, Ansible, Docker, HTOP.
- **Certifications:** Red Hat Certified System Administrator (RHCSA), LPIC-1.

4. Cloud Administrator

- **Responsibilities:** Configures cloud platform resources (AWS/Azure), provisions virtual servers, monitors costs, and manages access permissions.
- **Skills:** Cloud networking, access management policies, resource monitoring.
- **Tools:** AWS Console, Azure Portal, Terraform, AWS CLI.
- **Certifications:** AWS Certified SysOps Administrator, Azure Administrator (AZ-104).

Team Collaboration

In ApexByte Solutions Inc., these four positions support company operations together. The Helpdesk Technician handles day-to-day desktop support. Network issues are escalated to the Network Administrator. Local server issues are handled by the Linux Administrator, while external deployments and cloud services are managed by the Cloud Administrator.

PART 7: INFRASTRUCTURE RECOMMENDATIONS

1. **Internet Service Provider:** Subscribe to a Dedicated Business Fiber Plan (1 Gbps) with an uptime SLA and static IP to maintain constant connectivity for software development tasks.
2. **Server Hardware:** Use a dual-processor Intel Xeon server with 64GB ECC RAM and RAID 10 storage to protect against disk failures and RAM corruption.
3. **Backup Strategy (3-2-1 Rule):** Maintain 3 copies of data across 2 different media types (Local NAS and External Drives) with 1 encrypted copy stored offsite or in the cloud.

4. **Security Policies:** Deploy Microsoft Defender across all PCs, require 12+ character passwords with Multi-Factor Authentication (MFA), and enforce automatic screen locking.
5. **Expansion Planning:** Use standard /24 subnets and leave extra ports on the switch to accommodate future staff growth without changing core hardware.

PART 8: PERSONAL REFLECTION

Working on this Week 2 project was a real eye-opener for me. Before starting this, I always thought being a System Administrator was mostly about fixing broken computers, installing software, or troubleshooting network errors. But going through the process of building an entire IT infrastructure plan from scratch made me realize that proper planning and strategy are actually half the job.

The most challenging part of the project for me was figuring out the right hardware specs and setting up the network topology. I had to carefully think about what each department actually needed so I wouldn't waste budget on overpowered PCs for basic admin tasks, while still giving the IT dev team enough power for their tools and local testing environments. Designing the network diagram in Draw.io was also tricky because I had to make sure the firewalls, switches, and VLANs were arranged logically to keep the network secure and organized.

This project really showed me why planning before deployment is super important. If a company just buys hardware without a clear plan, they'll end up with incompatible equipment, random security holes, and a lot of wasted money. Having clean documentation means anyone on the IT team can easily step in, troubleshoot issues, or scale up the network when the company grows.

Overall, this project definitely pushed me to think more like an actual SysAdmin. It helped me get better at technical documentation, hardware inventorying, and network design—skills that I know will be useful as I keep building my portfolio and preparing for real-world IT work.